Health, Habits, and Cancer: Examining Risk of Cancer Using Support Vector Machines

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Introduction

- Cancer is second leading cause of death in the United States [1]
  - Common risk factors include alcohol and tobacco consumption, having a family history of cancer, and obesity [2]
- **Problem**: Can a medical history of cancer be predicted based on demographic and health data?
  - Factors of interest: age, income level relative to US national poverty level, body mass index (BMI), health insurance coverage status, frequency of alcohol consumption, duration of moderate physical activity (lasting longer than 10 minutes), usual hours of sleep per day
- **Approach**: Binary classification with support vector machines (SVMs) – linear, radial, polynomial kernels
  - Support vector machines are efficient for separating data into two or more classes
  - Kernels allow for faster computation and can help models capture linear and nonlinear relationships between predictors and target variable
- **Dataset**: 2022 National Health Interview Survey as accessed through IPUMS Health Survey [3]
  - 35,000+ initial observations
  - 7,571 observations used in modeling
    - Subset contained only individuals in the American South as well as individuals over the legal drinking age of 21
  - Demographic information, health metrics, health habits, history of disease

Theoretical Background

- SVMs find the optimal boundary (hyperplane) between two classes by maximizing the margin between this boundary and the nearest data points (support vectors)
  - Equation for hyperplane: $f(x) = \beta_0 + \Sigma\alpha_i < x_i, x_{i'} >$
- Size of margin controlled by cost parameter (C)
  - Small values of C allow for wider margins (more misclassifications; more bias, less variance)
  - Large values of C necessitate thinner margins (little to no misclassifications; less bias, more variance)
- Kernels allow for computing relationships between variables without enlarging feature space
  - Often aids in separating classes that cannot be separated linearly
  - Key hyperparameters: cost (all models/kernels), gamma (radial), degree (polynomial)
  - Linear kernel: $K(x_i, x_{i'}) = < x_i, x_{i'} >$
    - Forms a linear hyperplane without enlarging the feature space
  - Radial kernel: $K(x_i, x_{i'}) = exp\,(-\gamma\Sigma(< x_i, x_{i'} >)^2)$
    - Small gamma values limit sphere of influence, focusing predominantly on observations/data points that are close to one another
    - Large gamma values expand the sphere of influence, looking at data points that are far away from each other
  - Polynomial kernel: $K(x_i, x_{i'}) = (r + < x_i, x_{i'} >)^d$
    - Higher degree values result in more inflection points, or points where the function alters its direction
- Evaluation metrics: accuracy/error rate, ROC-AUC

Methodology

- Dataset reduced to 7,571 observations to only include individuals above the age of 21 from the American South
- Target variable: medical history of cancer (yes or no)
  - Variable recoded from values of 1 and 2 (no history of cancer vs. history of cancer) to binary values of 0 and 1
  - Variable showed significant class imbalance – individuals with no history of cancer account for ~88% of observations compared to 12% with a history of cancer
- Predictor variables: age, income level relative to US national poverty level, body mass index (BMI), health insurance coverage status, frequency of alcohol consumption, duration of moderate physical activity (lasting longer than 10 minutes), usual hours of sleep per day
- Sentinel values (numerical codes for non-responses) were replaced with null values and removed from dataset
- 80/20 train-test split followed by applying StandardScaler to training data to prevent dominance of single predictor variables
- Hyperparameters tested for each kernel:
  - Linear: Cost (C): [0.01, 0.1, 1, 5, 10]
  - Radial: Cost (C): [0.01, 0.1, 1, 5, 10], gamma (γ): [0.0001, 0.001, 0.01, 0.1, 1]
  - Polynomial: Cost (C): [0.01, 0.1, 1, 5, 10], degree (d): [2, 3]
- 'class_weight' parameter for each model set to 'balanced' to account for class imbalance in target variable
- Scoring metric for models:
  - ROC-AUC used within model to account for both class imbalance and provide more reliable information than accuracy
  - Accuracy/error rate to assess model performance on both training and test data
  - All metrics applied on both untuned and tuned models

Results

Model Comparisons

Kernel Type	Training Error Rate (pre-tuning)	Test Error Rate (pre-tuning)	Training Error Rate (post-tuning)	Test Error Rate (post-tuning)	Train AUC Score	Test AUC Score	Optimal Hyperparameters
Linear	35.04%	34.39%	35.02%	34.39%	0.79	0.79	cost = 10
Radial	33.70%	34.65%	35.40%	34.92%	0.79	0.80	cost = 10; gamma = 0.001
Polynomial	27.03%	27.72%	35.54%	35.58%	0.79	0.78	cost = 10; degree = 2

Figure 1. Table of error rates, AUC scores, and optimal hyperparameters for each support vector machine model

- Radial kernel generalized best with unseen data, with test AUC score of 0.80, followed by linear and polynomial models
- Tuned models generated roughly equivalent training and test error rates

MOD10DMIN	-0.028647
BMICALC	-0.023493
ALCANYNO	0.003788
POVERTY	0.026453
HRSLEEP	0.037802
HINOTCOVE	0.105317
AGE	0.336127
CANCER	1.000000

Figure 2. Correlation coefficients for each predictor variable relative to target variable (CANCER)

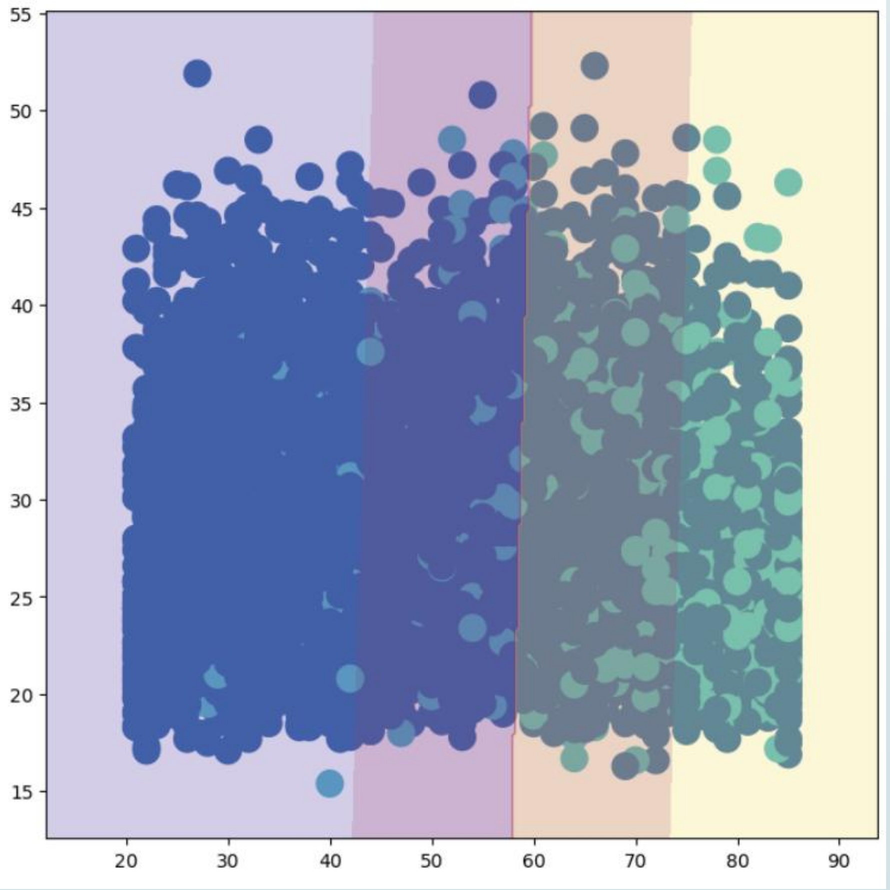

Figure 3. Linear decision boundary for linear kernel model comparing age (x-axis) and body mass index (y-axis)

Discussion

- All tuned models achieved similar AUC scores and training and test error rates
  - Linear kernel model was most stable between tuned and untuned models, suggesting the underlying relationship between history of cancer and predictors is linear
  - Selected predictors may be limited
- Polynomial kernel model lowered accuracy following tuning, suggesting potential overfitting to the training data
- Age and health insurance coverage status have strongest correlative relationship with target variable among chosen predictors

Conclusion

- All three SVM models performed well with predicting cancer class, with AUC scores at or around 0.80
- Similar model performance on test data suggests limitation with chosen predictors
  - Future research may investigate other factors such as tobacco usage or diet
  - Age shown to be strongest predictor among those chosen, with individuals of older age being more likely to have or have had cancer
- Results suggest that more rigorous cancer screening as one ages is necessary and justified, given that older populations are most at risk relative to their younger counterparts
- Health insurance coverage status as second-highest correlative predictor suggests increased coverage improves screening
  - Positive correlation coefficient suggests those with health insurance are more likely to be diagnosed
  - Expansion of healthcare coverage could provide more accurate data with respect to cancer rates, especially in poorer and uninsured populations

References

[1] Centers for Disease Control and Prevention. (2024, June 3). *Cancer*. Chronic Disease Indicators. https://www.cdc.gov/cdi/indicator-definitions/cancer.html

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[4] PosterNerd. (n.d.). *Scientific poster PowerPoint templates – Conceptualizing Cobalt*. https://www.posternerd.com/sciposters-templates?signs_redirect